

Problem 8.8

An investor would like to have \$10,000 at the end of 5 years. The annual simple rate of discount is 3%. How much should the investor deposit today to reach that goal?

Solution

Under simple discount, the present value of a future amount F due in t years at discount rate d is

$$P = F(1 - dt).$$

Here,

$$F = 10,000, \quad d = 0.03, \quad t = 5.$$

So,

$$P = 10,000(1 - 0.03 \cdot 5) = 10,000(1 - 0.15).$$

Thus,

$$P = 10,000(0.85) = 8,500.$$

Therefore, the investor should deposit

$$\boxed{\$8,500}.$$